Final Project 2

Multi-Image Creative Generation

(Choose ONE Task: Story or Virtual Try-On)

Course: DSAA2012
Instructor: Prof. Chen Liang

Overview

In this project, your team will build a system that generates **multiple images** that must be logically connected and mutually consistent.

You must choose **one** of the following two task options:

1. **Task 1: Story (Multi-Shot / Multi-Panel Generation)**
   Generate a sequence of images from a multi-panel text description.
2. **Task 2: Virtual Try-On (Style-Consistent and/or Multi-View Generation)**
   Generate try-on image(s) from a person image and a garment image.

Important: This is a **choose-one-of-two** project. Each team should complete **either Task 1 or Task 2**, not both.

Provided Materials

The TA team will provide:

- a released development test set (**Test-A**),
- a hidden holdout test set (**Test-B**) for grading,
- input and output format specifications,
- a submission template,
- starter code for loading inputs and packaging outputs.

Data collection is part of the project. Each team must **find, select, and prepare its own training/finetuning data** (if needed) for the chosen task. You may use public datasets, self-collected data, or other permissible external resources, provided that they comply with course policy and are fully disclosed in your report.

Important Rules

This project has strict constraints:

- You may use **Test-A** for development and reporting only.
- Your final system must generalize to the hidden **Test-B**.
- **Manual per-case editing is not allowed.**
- **Hard-coding outputs for specific test cases is not allowed.**
- **Agent-based systems are not allowed and no external API usage is allowed.**
- Each team is responsible for obtaining its own training data or external resources.

- Any external resources (pretrained models, public datasets, third-party tools, etc.) must comply with course policy and must be clearly disclosed.

Your submission must be a **fixed, automatic, and reproducible pipeline**.

Task 1: Story

Goal

Given a sequence of text descriptions (shots/panels), generate a corresponding sequence of images.

Your outputs should satisfy

- **Per-panel correctness:** each image matches its panel description.
- **Cross-panel consistency:** recurring characters, objects, background, and style remain consistent across the sequence.
- **Narrative continuity:** the story progresses naturally from one panel to the next.

Output

- Generate the required number of images for each test case.
- Follow the official file naming and resolution requirements.

Task 2: Virtual Try-On

Goal

Given a person image and a garment image, generate try-on result(s) that preserve the person's identity and faithfully reflect the garment.

Depending on the test case, you may need to output:

- a single try-on image, or
- multiple try-on images with consistent identity, garment appearance, and style across views or poses.

Output

- Generate the required number of images for each test case.
- Follow the official file naming and resolution requirements.

Evaluation

Your project will be evaluated **primarily on the hidden Test-B**. Each team will be evaluated **only on the task it chooses**.

Grading will be based mainly on **human evaluation by the TAs and the instructor**. We care about the **relative quality of submissions**, not an absolute automatic score. In other words, your project will be judged by how well it performs **compared with other teams' submissions** under the same task setting.

Task 1: Story — evaluation focus

- prompt adherence for each panel,
- narrative continuity,
- cross-panel consistency,
- style consistency,
- image quality.

Figure 1: Sample image, for illustration of the task format only; it is not a standard answer and does not constitute any reference for evaluation.

Task 2: Virtual Try-On — evaluation focus

- garment fidelity,
- identity preservation,
- natural garment-body integration,
- cross-image consistency (if applicable),
- aesthetic quality and visual completeness.

Figure 2: Sample image, for illustration of the task format only; it is not a standard answer and does not constitute any reference for evaluation.

Bonus consideration: Bonus consideration: For Virtual Try-On, multi-view generation is not required. However, if a submission includes strong multi-view results with good consistency in identity, garment details, and overall style, it may receive a higher base score before the final evaluation is conducted.

Report Requirements

Your technical report must include:

- a clear description of your full system,
- all major decision steps used to generate outputs,
- controlled comparisons on **Test-A** to justify design choices,
- **8–12 qualitative examples** for your chosen task,
- brief failure analysis,
- a description of the datasets you used, including how they were obtained, selected, filtered, or processed,
- a **compliance statement** confirming:

  - full automation,
  - no manual per-case editing,
  - no hard-coding,
  - no agent-based system,
  - full disclosure of external resources.

Submission Requirements

Each team must submit:

- runnable code,
- a README,
- output files in the official format for the chosen task,
- a technical report,
- any other materials required by the course staff.

Submissions must be reproducible and runnable by the TA team.

Academic Integrity

The following are prohibited:

- manual editing for specific test cases,
- hard-coding outputs,
- using agent-based systems,
- exploiting hidden test information,
- concealing external resources or datasets,
- fabricating results.

Violations will be handled according to course academic integrity policy.

Summary

This project evaluates your ability to build a **stable, automatic, and generalizable multi-image generation system** under strict constraints.

Choose **one** task:

- **Story**, or
- **Virtual Try-On**.

A strong submission should produce outputs that are:

- faithful to the input,
- consistent across images,
- visually convincing,
- and generated by a fair, reproducible pipeline.